Do Now · retrieval

Linear search: trace and code

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What property must both inputs satisfy before a standard merge step?

Q2

What does the key represent in insertion sort?

Q3

What is the last index of a 5-element C# array?

Q4

A loop checks a[0], a[1], a[2] then returns. How many comparisons were made?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Begin at index zero — Linear search examines one value at a time, in order.

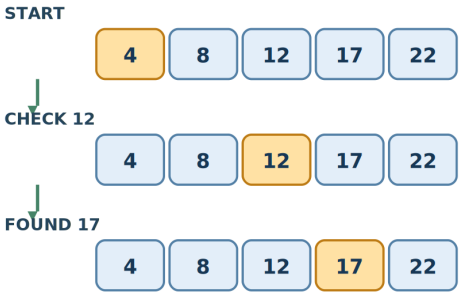
Core idea 2

Stop at the first match — When a[i] equals target, return i immediately.

Core idea 3

No match — After scanning every element, return -1; no ordering assumption is needed.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

target — Value the search seeks.

index — Zero-based position of an array element.

sentinel — Special result such as -1 meaning not found.

early return — Leave the function as soon as a match is found.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

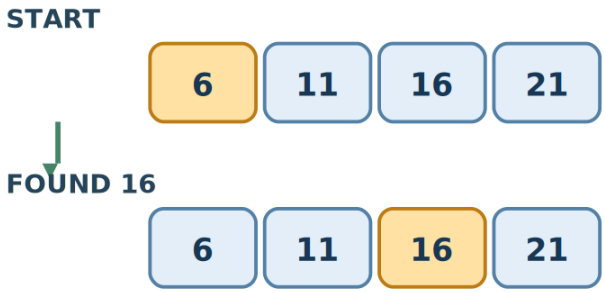
Problem

Find 16 in [6,11,16,21]. Give the index and number of comparisons.

Follow the worked method

1. Predict first — Write visited indexes before identifying the match.
2. Model solution — Visit 0 (6), 1 (11), 2 (16). Return index 2 after three comparisons.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
Which indexes will be visited?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Search for 7 in [4,7,9,12]. Give index and comparison count.

Question 2

Search for 5 in [3,8,11,16]. Give return value and count.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Modify the function to count comparisons without returning early; explain the output for a 5-element array.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Linear search requires sorted input.

Use key vocabulary from memory

1. Explain target in your own words.

2. Explain index in your own words.

3. Explain sentinel in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.